The R Ecosystem

An Architectural, Statistical, and Practical Overview

From Lexical Scoping to Production-Grade Graphics

Abate Alemayehu

Ethiopian Statistics Service (ESS)

September 2, 2026

Agenda & Learning Outcomes

Foundations

- The S lineage and R's philosophical roots.
- Lexical scoping & lazy evaluation.
- Object-oriented systems (S3, S4, RC).

The Ecosystem

- CRAN vs. Bioconductor.
- The Tidyverse paradigm.
- RMarkdown / Quarto for reproducibility.

Statistical Superpowers

- Linear models to GAMs.
- Time-series (ARIMA, Prophet).
- Machine learning (caret, tidymodels).

Production Workflows

- RStudio IDE & Project management.
- Shiny for interactive dashboards.
- R in DevOps (plumber APIs).

Defining R: More Than Just a Language

The Official Definition

R is an open-source, interpreted programming language and integrated software environment for statistical computation, data manipulation, and high-performance graphics.

- **Language:** Functional, weakly object-oriented, dynamically typed.
- **Environment:** Includes a console, debugger, graphics device, and a robust package system.
- **Philosophy:** Built by statisticians, *for* statisticians — not by computer scientists.

Key Paradigm

R treats **data as first-class citizens**. Every operation is vectorized; loops are often abstracted away in favor of functional primitives (`lapply`, `purrr`).

The Genesis: From Bell Labs to Global Standard



- 1976: Bell Labs develops **S** for interactive data analysis.
- 1993: Ross Ihaka & Robert Gentleman release R (the "S" successor) at the University of Auckland.
- 1997: The R Core Team is formed; CRAN is launched.
- 2000: R 1.0.0 released — stable enough for production.

Why "R"?

A tribute to the S language and the initials of its creators (**R**oss & **R**obert).

Key Evolution: R has moved from academic niche to enterprise-grade, now powering analyses at **Google**, **Facebook**, and **Merck**.

Under the Hood: Lexical (Static) Scoping

What is it?

- R determines the value of a variable based on where the **function is defined**, not where it is called.
- Enables **closures** — functions that carry their defining environment.

```r
make_counter <- function(init = 0)
    {
  count <- init
  function() {
    count <<- count + 1
    count
  }
}
counter_a <- make_counter(10)
counter_a() # 11
counter_a() # 12
# Environment is preserved!
```

Why this matters

This is the backbone of function factories and statistical model objects (e.g., `lm()`) which store their call and data environments for later `summary()` and `predict()` methods.

Object-Oriented Systems: The "Three Pillars"

System	Style	Key Use Case
S3	Generic-function OOP	90% of base R and CRAN. Simple, informal. e.g., `print.lm()`
S4	Formal OOP	Bioconductor, complex genomics. Strict validation, multiple inheritance.
RC (R6)	Reference Classes	Mutable objects, stateful behavior. Used in **Shiny** and **plumber**.

S3 Example (The Most Common)

```
x <- rnorm(100)
class(x) <- "my_dist"
print.my_dist <- function(obj) cat("Custom distribution with mean:",
    mean(obj))
print(x) # Calls our custom method!
```

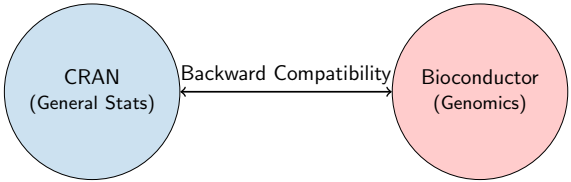
The Power of Extensibility: CRAN + Bioconductor

Comprehensive R Archive Network (CRAN)

- **20,000+** packages.
- Strict compilation, testing, and documentation policies.
- Every package undergoes automated daily checks.

Bioconductor

- Specialized for **genomics, proteomics, and bioinformatics**.
- Uses a 6-month release cycle for version stability.
- Hosts `limma`, `DESeq2`, `edgeR`.



Installation: `install.packages("dplyr")` (CRAN) vs `BiocManager::install("DESeq2")` (BioC).

The Tidyverse: A Coherent Grammar of Data

Core Philosophy (Hadley Wickham)

"Tidy datasets are easy to manipulate, model, and visualize. Each variable is a column, each observation is a row, and each type of observational unit is a table."

Core Packages:

- `ggplot2` — Grammar of Graphics.
- `dplyr` — Data manipulation (verbs: `select`, `filter`, `mutate`, `summarise`).
- `tidyr` — Reshaping (pivot_longer, pivot_wider).
- `purrr` — Functional programming.

```
1  library(tidyverse)
2
3  iris %>%
4  group_by(Species) %>%
5  summarise(avg_length = mean(
       Sepal.Length)) %>%
6  ggplot(aes(x = Species, y =
       avg_length)) +
7  geom_col(fill = "steelblue")
```

%¿% = The "pipe" operator (forward-pipe).

R Markdown: The Gold Standard for Reproducibility

- An **executable notebook** format that combines code, output, and narrative in a single document.
- Supports multiple outputs: PDF, HTML, Word, and even **Beamer** presentations!

Basic YAML Header

```
---
title: "My Analysis"
author: "Your Name"
output: html_document
---
```

- **Code Chunks:** Embed R code inline or in blocks.
- **Caching:** Store expensive computations via `cache=TRUE`.
- **Parametric Reports:** Render the same report with different input parameters (`params:`).

This paradigm has been extended by **Quarto**, the next-generation multi-language scientific publishing system.

Statistical Modeling: Depth and Breadth

- **Linear Models:** `lm(y ~ x1 + x2)` — the workhorse of econometrics.
- **Generalized Linear Models (GLM):** `glm(..., family = binomial())` for classification.
- **Generalized Additive Models (GAM):** `mgcv::gam(y ~ s(x))` — non-parametric smoothing.
- **Mixed-Effects:** `lme4::lmer(y ~ x + (1|group))` for hierarchical data.

Model	R Package	Application
OLS / Ridge	`lm`, `glmnet`	Economics / Marketing
Logistic Regression	`glm`	Medical Diagnostics
ARIMA / GARCH	`forecast`, `rugarch`	Financial Time Series
Survival Analysis	`survival`	Clinical Trials

The Grammar of Graphics: ggplot2

Core Layers:

1. **Data** — The source.
2. **Aesthetics (aes)** — What maps to x, y, color, size.
3. **Geoms** — How to plot (points, lines, bars).
4. **Stats** — Transformations (smoothers, binning).
5. **Scales** — Axes, colors, themes.

```
ggplot(mtcars, aes(x=wt, y=mpg
    , color=factor(cyl))) +
geom_point(size=3, alpha=0.7)
    +
geom_smooth(method="lm", se=
    FALSE) +
scale_color_brewer(palette="
    Set1") +
theme_minimal() +
labs(title="Weight vs. Mileage
    ",
x="Weight (1000 lbs)")
```

Extensibility

Extensions like `ggridges`, `ggrepel`, and `plotly::ggplotly()` elevate static plots to interactive dashboards.

R vs. Python: When to Choose Which?

Criterion	R	Python
Primary Domain	Statistics & Research	General Purpose & ML
Data Ecosystem	Tidyverse (cohesive)	Pandas (flexible)
Visualization	`ggplot2` (superior)	Matplotlib/Seaborn
Deep Learning	Limited (Keras wrapper)	First-class (PyTorch/TF)
EDA Speed	Extremely fast (interactive)	Moderate
Production/API	`plumber`	`FastAPI`, `Flask`

The Modern Data Scientist

Many workflows are **bilingual**: Explore and model in R (for its statistical rigor), then deploy in Python (for software engineering).

RStudio (Posit): Beyond a Text Editor

Four-Pane Layout:

1. **Source Editor** — Scripts, RMD, functions.
2. **Console** — Interactive execution.
3. **Environment/History** — View objects, history.
4. **Files/Plots/Packages/Help** — All peripheral tools.

Advanced Features:

- **Projects (.Rproj)**: Isolate libraries and working directories.
- **Addins**: UI tools for styling, citations, snippets.
- **Debugging**: Breakpoints, traceback, and `browser()` integration.
- **Version Control**: Native Git integration.

Posit Workbench extends this to enterprise with multi-session, administrator control, and visual analytics.

Shiny: Turning Analyses into Web Apps

Core Concept

Shiny is an R package that uses a reactive programming model to build interactive web applications entirely in R — no HTML/CSS/JS required (unless you want to customize!).

UI (User Interface):

- `fluidPage()`, `sidebarLayout()`
- Inputs: `sliderInput()`, `selectInput()`
- Outputs: `plotOutput()`, `tableOutput()`

Server (Reactivity):

```
server <- function(input, output) {
  output$distPlot <- renderPlot({
    hist(rnorm(input$obs),
    col = 'darkgray')
  })
}
shinyApp(ui, server)
```

Shiny Dashboards can be hosted on **shinyapps.io**, Posit Connect, or even embedded in RMarkdown documents.

R in Production: APIs with `plumber`

What is Plumber?

- Transforms R functions into RESTful APIs.
- Uses Roxygen-style comments to define endpoints.
- Supports JSON, serialization, and HTTP verbs.

Deployment Options:

- Dockerize your R API.
- Deploy on AWS EC2, Google Cloud, or Azure.
- Scale using `parallel` or `future` packages.

```r
#* @get /predict
function(input_val) {
  model <- readRDS("model.rds")
  predict(model, newdata = data.frame(x = input_val))
}
# Run with: pr() %>% pr_run(port=8000)
```

This bridges the gap between data science and software engineering.

Your First 24 Hours with R

1. **Install R** from CRAN (cran.r-project.org).
2. **Install RStudio** from Posit's website (Free Desktop version).
3. **Explore the Console:** Type `2 + 2`, then `?mean`.
4. **Install tidyverse:** `install.packages("tidyverse")`.
5. **Load and inspect:** `library(tidyverse); glimpse(iris)`.
6. **Build your first plot:** `ggplot(iris, aes(x=Sepal.Length, y=Petal.Length)) + geom_point()`.

Pro Tip

Use the built-in `demo()` function! Try `demo(graphics)` or `demo(lm.glm)` to see R's capabilities in real-time.

The R Community: Where to Learn and Ask

Books

- *R for Data Science* (Wickham, & Grolemund) — Free online.
- *Advanced R* (Wickham) — For deep understanding.
- *The R Cookbook* (Long & Teetor) — Practical recipes.

Online Communities

- Stack Overflow (tag: R)
- RStudio Community
- Reddit (r/rstats)

Conferences & Events

- **useR!** — The international R user conference.
- **RStudio Conf** (now Posit Conf).
- Local R User Groups (Meetup).

Newsletter

- **R Weekly** — A curated digest of R news, packages, and tutorials.

References I

- Wickham, H., & Grolemund, G. (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media.
- Wickham, H. (2019). *Advanced R* (2nd ed.). Chapman and Hall/CRC.
- Chambers, J. M. (2016). *Extending R*. Chapman and Hall/CRC.
- Ihaka, R., & Gentleman, R. (1996). R: A Language for Data Analysis and Graphics. *Journal of Computational and Graphical Statistics*, 5(3), 299–314.
- Posit Team. (2023). *Shiny: Web Application Framework for R*. shiny.posit.co
- R Core Team. (2024). *The Comprehensive R Archive Network*. cran.r-project.org

Thank You

Questions?

- **Email:** abatea72@gmail.com
- **Slides available at:** `https://github.com/abatea/r-presentation`
- **Follow-up reading:** *R for Data Science* (free online)

"The simple graph has brought more information to the data analyst's mind than any other device." — John Tukey